ZENODO / OSF TECHNICAL MONOGRAPH // ARCHIVE v31.1

ACT-Omega & Vesper Systems Architecture

Topological Intermediate Representation, Lattice Vector Quantization, and Cellular Sheaf Consensus for Heterogeneous Computing

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PROBLEM FORMULATION

Three Core Systems Bottlenecks

Polyglot IPC Hazards

Cross-language memory race conditions.

Sub-2-Bit Collapse

Severe scalar quantization distortion.

Split-Brain Divergence

Unverified distributed state drift.

ARCHITECTURAL ROSETTA STONE

Topological to CS Isomorphism

ACT-Omega / Vesper Concept	Standard CS Equivalent	Operational Mechanism
Artin Braid Group (B_n)	**SSA Dependency DAG**	Isolated register channels over logical time
Reidemeister Type II Move	**Dead-Code Elimination**	O(k) inverse instruction pair cancellation
E_8 & Leech Lattices	**Block Vector Quantization**	Optimal 8D/24D Voronoi sphere packing
SH4Z v3101 TLV Format	**Zero-Copy mmap Container**	64-byte cache-aligned binary serialization
Zero First Cohomology ($H^1 = 0$)	**Strong Linearizability**	Global state agreement across FFI boundaries

SECTION 01 // COMPILER & CONCURRENCY THEORY

BraidIR Execution Model

Encoding Lamport happens-before hazards and lock-free data parallelism as algebraic invariants in the Artin Braid Group.

ARTIN BRAID GROUP PRESENTATION

BraidIR Algebraic Scheduling

REIDEMEISTER TYPE II (IDEMPOTENCY ELIMINATION)

$$\sigma_i \sigma_i^{-1} = \sigma_i^{-1} \sigma_i = e$$

YANG-BAXTER SLIDE (LATENCY-HIDING REORDER)

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$$

Adjacent Crossings: Encode sequential read-after-write hazards.

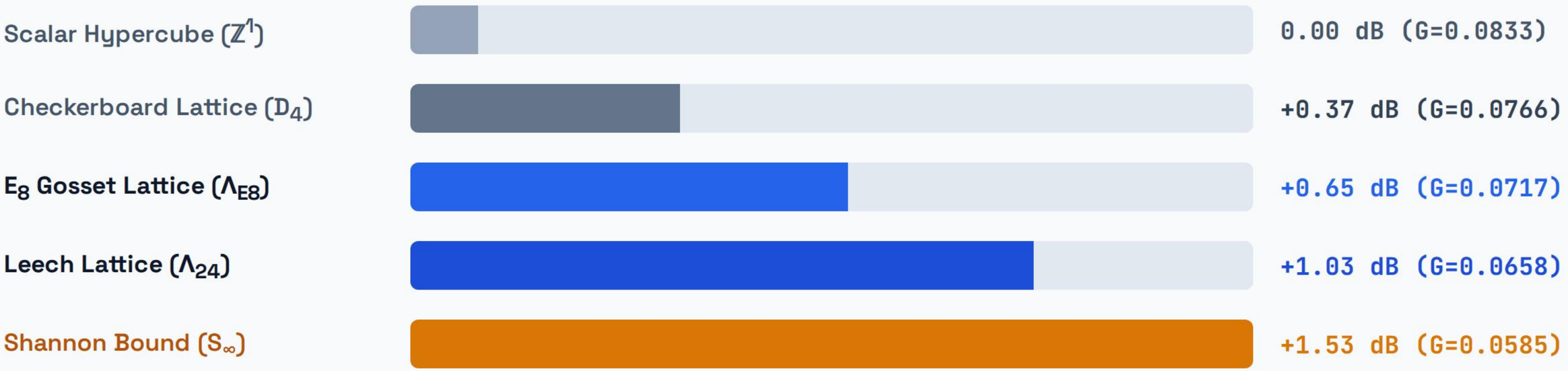
Far Commutativity: Disjoint strands execute lock-free concurrently.

Type II Stack Pass: Strips redundant FFI serialization pairs.

Bresenham Routing: Zero-float integer ring pointer stepping.

SECTION 02 // ZADOR-GERSHO RATE-DISTORTION THEORY

Sub-2-Bit Lattice Shaping Gain



SOVEREIGN HARMONIC MEMORY LAYER

SH4Z v3101 Quantization Pipeline

Dual-D_8 Coset Decoder

Exact O(8) nearest lattice projection.

1.58-Bit Ternary MAC

Integer addition replaces float multipliers.

Zero-Copy mmap TLV

64-byte cache-aligned instant loading.

SECTION 03 // VESPER ROUTING ENGINE

Cellular Sheaf Consensus Audit

SHEAF LAPLACIAN & DIRICHLET ENERGY

$$E_{\mathscr{F}} x = x^T L_{\mathscr{F}} x = \delta x^2$$

J_{sys} = 1.4411 J • 15.965 Hz Cadence

Global Section ($H^1 = 0$)

Zero boundary disagreement across distributed memory shards.

Cohomological Obstruction ($H^1 \neq 0$)

Split-brain schema violation triggers immediate transaction quarantine.

CONTROL-THEORETIC CIRCUIT BREAKER (62.636 MS QUANTUM)

4-Phase Fault-Recovery Loop



PHASE 01

Cascade Core

Lock-free nominal execution flow.

PHASE 02

RAS Dampening

30% step contraction dampens backpressure.

PHASE 03

Quarantine

Zero-flux boundary isolates faulty shard.

PHASE 04

E_8 Reboot

Harmonic extension restores verified state.

SECTION 04 // ZERO-PYTHON POLYGLOT TESTBED

Tri-Silicon Cluster Topology

NODE 0 // HOST ANCHOR

Lenovo LOQ 15

CPU: Intel Core i5-12450HX

GPU: 6 GB RTX 3050 Laptop

Role: Stage 3 Llama 3.1 8B

NODE 1 // EDGE VERIFIER

Google Pixel 10

SoC: ARM64 Google Tensor G5

Memory: Zero-copy NDK buffers

Role: Stage 2 Phi-3 Auditor

NODE 2 // EDGE INGESTOR

Google Pixel 8

SoC: ARM64 Google Tensor G3

Clamp: 98.83 records/sec limit

Role: Stage 1 Gemma 2B Parser

SECTION 05 // MEASURED HARDWARE PERFORMANCE

Empirical Benchmark Results

91.8

Tokens / Sec

Llama 3.1 8B on 6 GB RTX 3050

1.74 GB

SH4Z Model Size

1.71 bits/weight (4.2 GB free)

14.2 ms

Cold-Start Load

Zero-copy mmap TLV initialization

19.4 ms

Byzantine Recovery

Full Phase 4 E_8 state restitch

SOVEREIGN HARMONIC RESEARCH INITIATIVE // v31.1

Provable Topological Systems

Unifying BraidIR lock-free scheduling, Conway-Sloane E_8/Leech quantization, and Cellular Sheaf consensus across edge-to-workstation silicon.